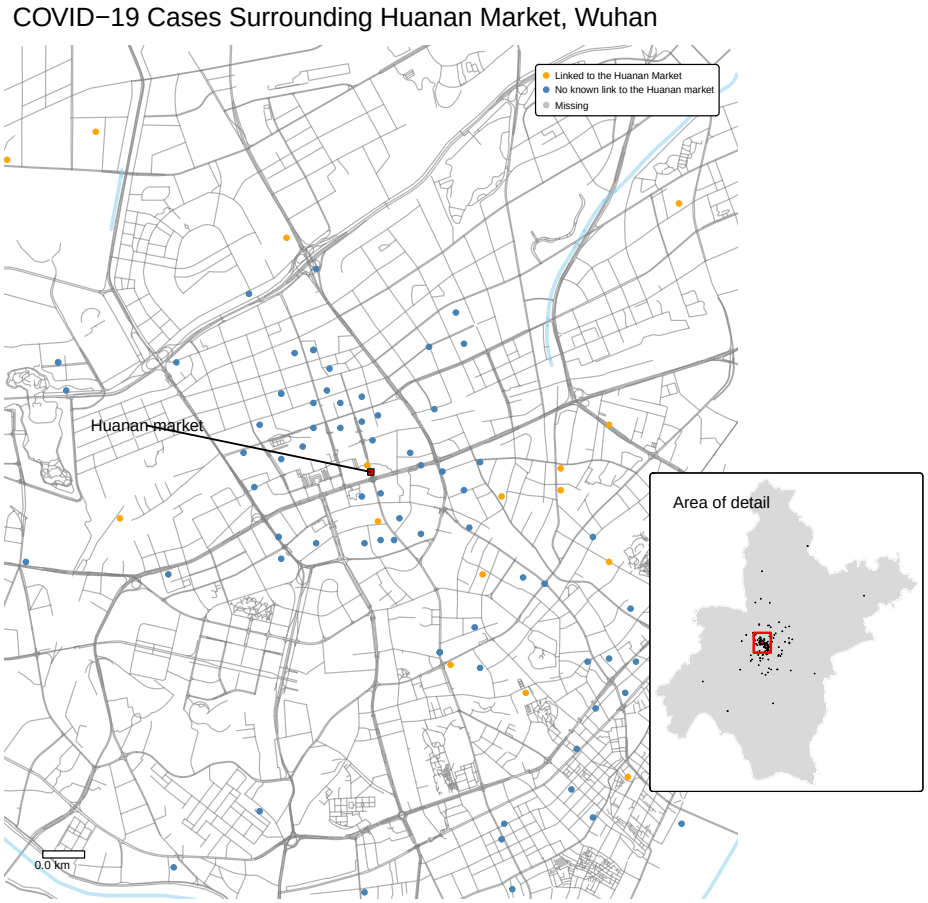
Data6500_A2_Treasure

Question 1

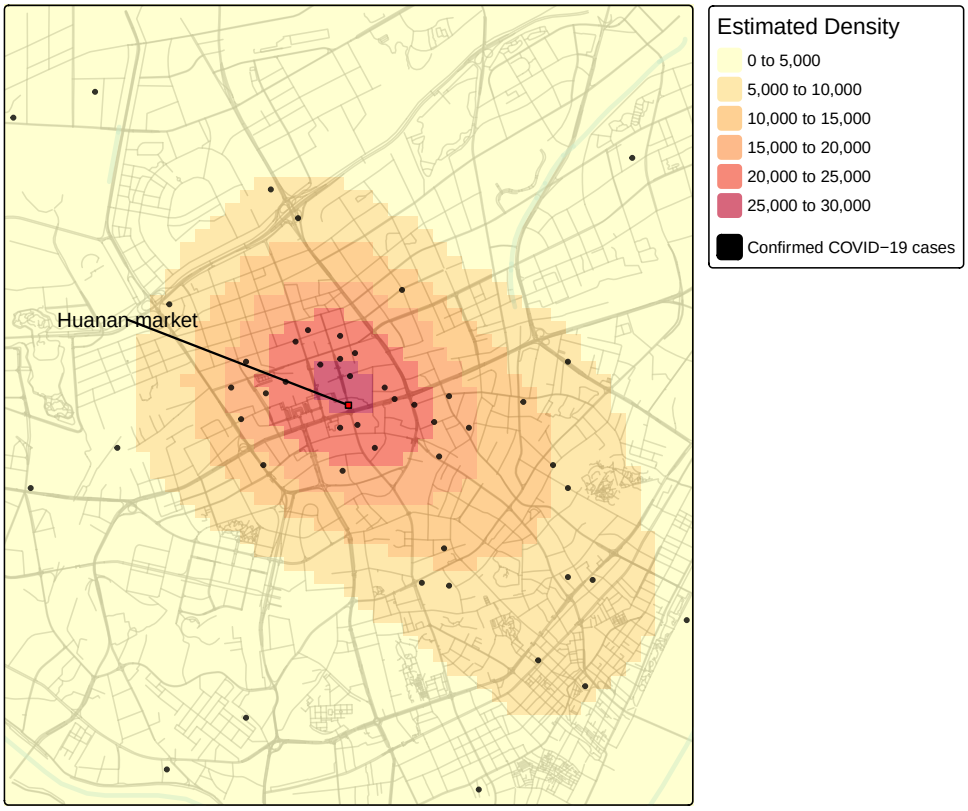
Part A

Plotting the graph of Wuhan's Human Seafood Wholesale Market



Part B

Determining if the Huanan Market was a COVID-19 hot spot using KDE

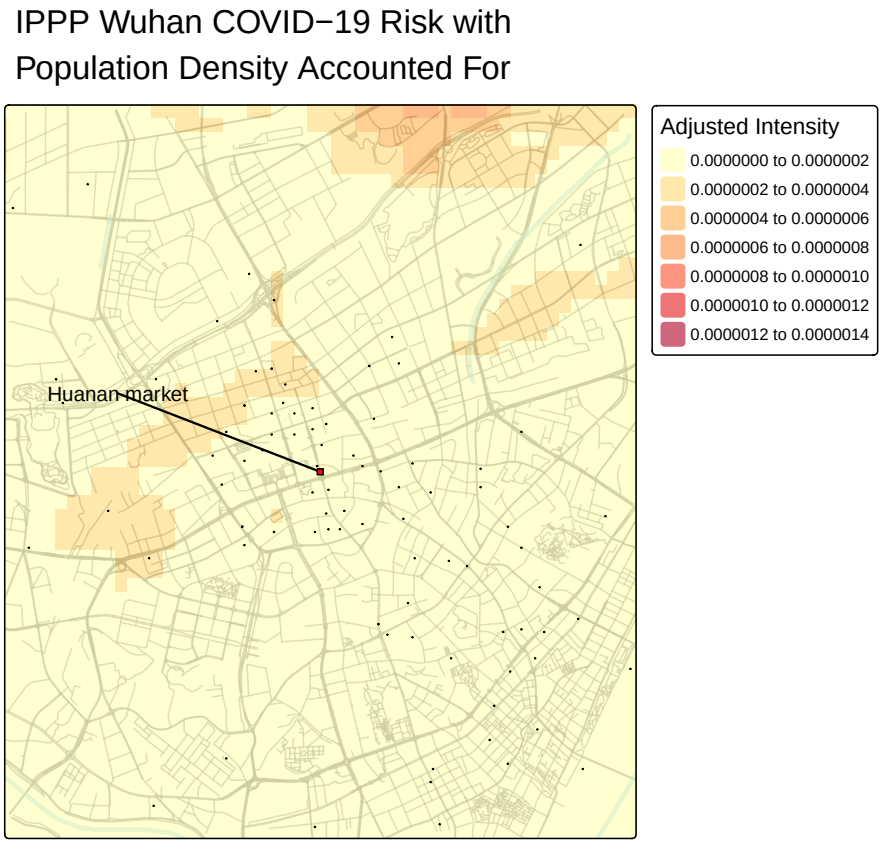


Based on the kernel density estimate (KDE) of confirmed COVID-19 cases, the Huanan Market appears to have been a significant hotspot for COVID-19 transmission. The market, represented by the red square, is situated within the highest-density region on the map, indicating a concentration of cases surrounding it. The spatial distribution pattern from the KDE suggests that the Huanan Market was a primary area of elevated transmission risk in the early outbreak phase.

Part C

While Part B suggests that the Huanan Market was a COVID-19 hotspot, this analysis does not account for the population density in Wuhan. To assess whether the market was truly a high-risk area when controlling for population distribution, an Inhomogeneous Poisson Point Process (IPPP) model will be applied. The IPPP model is well-suited for evaluating how case intensity varies spatially as a function of covariates, in this case, population density.

Using population as a spatial covariate, the model will help determine whether the observed clustering near the market reflects genuine elevated risk or simply areas of higher population concentration.



After adjusting for population density using the IPPP model, the results indicate a relatively low case intensity surrounding the Huanan Market. This suggests that the apparent clustering of cases observed in Part B may be largely attributable to the high population density or mobility in that area, rather than elevated transmission risk associated with the market itself. Therefore, the concentration of cases near the market could reflect where people live or move, rather than the market being a hotspot for COVID-19 transmission.

Question 2

Analysis of soil pH levels in Indiana. Here the goal is to determine the counties in Indiana that have suitable soil profiles for agriculture.

Since the provided soil dataset contains pH measurements from locations worldwide, it was necessary to isolate the data specific to Indiana for this analysis. To achieve this, the dataset

was first filtered to include only records from the B horizon, as this horizon is the focus of the study. Based on a review of soil classification practices and the depth information available in the dataset, a new column was created to categorize the B horizon data into B1, B2, and B3 sublayers using the upper depth values. This approach ensured that sublayers were systematically included in the analysis and facilitated clearer visualisation. Finally, using spatial coordinates, the soil data was spatially joined to Indiana county boundaries, retaining only the soil measurements located within the state.

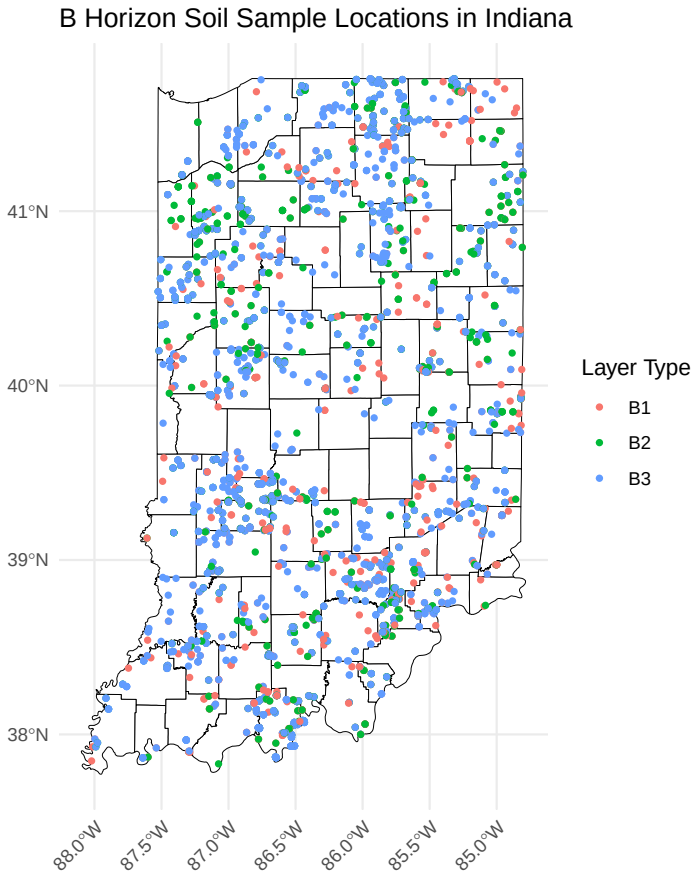
```
#filtering Horizon B soil information
#want the soil information for the layers that start with B
b_soil <- soil_info %>%
  filter(grepl("^B", layer_name))
```

```
#cleaning the B layer using the upper depth to determine what B Horizon each reading is
#eg: B1 <= 30, 30 < B2 <= 60, B3>60, everything else=other
b_soil <- b_soil %>%
  mutate(
    # Categorize into B1, B2, B3 based on upper depth
    depth_layer = case_when(
      upper_depth <= 30                    ~ "B1",
      upper_depth > 30 & upper_depth <= 60 ~ "B2",
      upper_depth > 60                     ~ "B3",
      TRUE                                 ~ "other"
    )
  )

#making the depth layers a factor
b_soil$depth_layer <- as.factor(b_soil$depth_layer)
levels(b_soil$depth_layer)
```

```
[1] "B1" "B2" "B3"
```

```
#head(b_soil)
```



Analysis

Kriging was performed to identify counties with ideal soil pH levels between 6.0 and 7.5. For this analysis, block kriging was used, as the objective is to estimate average soil pH over entire counties, rather than at individual point locations. Block kriging is better suited for this purpose because it provides predictions over spatial areas (blocks), allowing for assessment of overall soil suitability at the county level.

Kriging was performed separately for each sublayer to account for the fact that different crops typically utilize different soil depths. Stratifying the analysis by sublayer allows for a clearer assessment of pH suitability within each depth horizon, supporting more targeted crop management based on sublayer-specific soil conditions.

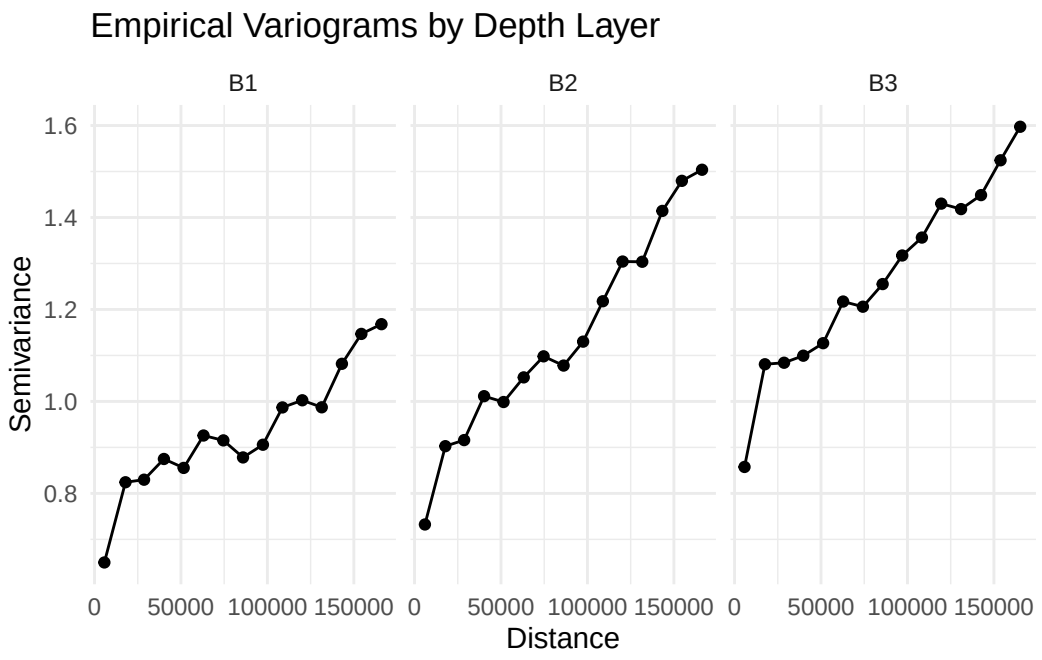
```
#B1 Variogram
v_B1 <- variogram(value_avg ~ 1, soil_sp_B1)
v_B1$Depth <- "B1"
model_B1 <- fit.variogram(v_B1, vgm(model = "Exp"))
```

```r
#B2 Variogram
v_B2 <- variogram(value_avg ~ 1, soil_sp_B2)
v_B2$Depth <- "B2"
model_B2 <- fit.variogram(v_B2, vgm(model = "Exp"))

#B3 Variogram
v_B3 <- variogram(value_avg ~ 1, soil_sp_B3)
v_B3$Depth <- "B3"
model_B3 <- fit.variogram(v_B3, vgm(model = "Exp"))

v_all <- bind_rows(v_B1, v_B2, v_B3)

#plotting all the of variograms for each B Horizon
ggplot(v_all, aes(x = dist, y = gamma)) +
  geom_point() +
  geom_line() +  # optional, to connect points
  facet_wrap(~Depth) +
  labs(x = "Distance", y = "Semivariance",
       title = "Empirical Variograms by Depth Layer") +
  theme_minimal()
```



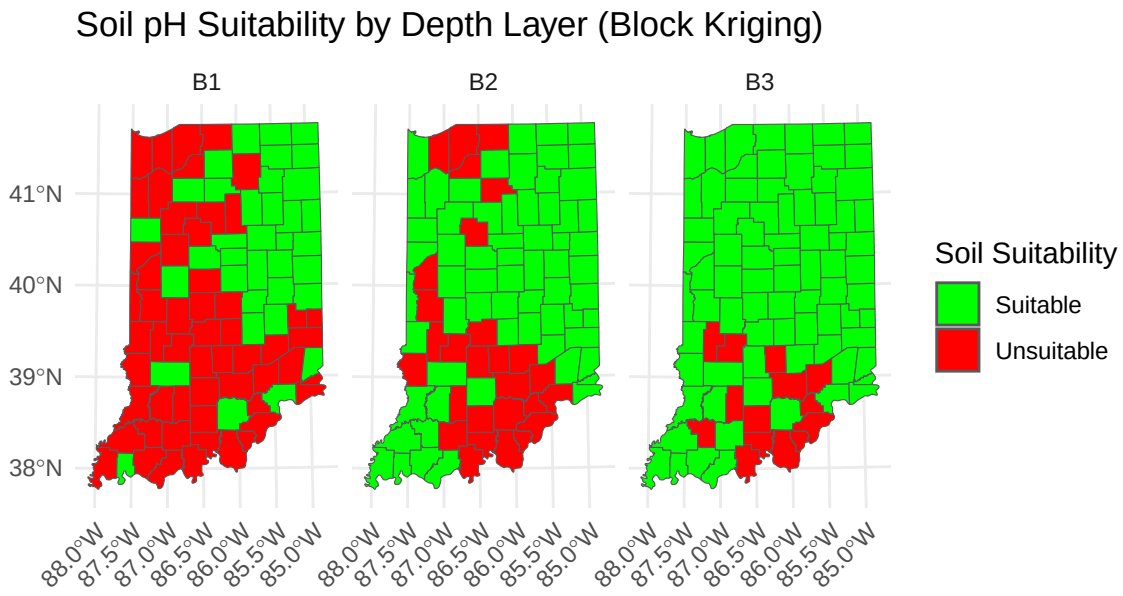
The variograms show that soil pH variability increases with distance in all three sublayers

(B1, B2, and B3), indicating spatial autocorrelation. The B1 layer shows lower variability, meaning pH values are more consistent at shallower depths, while the B2 and B3 layers show greater variability, suggesting deeper soils are less spatially consistent. This confirms that kriging is appropriate for this analysis, as the presence of spatial structure allows for reliable interpolation. The fitted variogram models will now be used to inform the kriging predictions for each sublayer.

```
#kiriging for each sublayer
#B1 Block Kriging
kriging_B1 <- krige(
  value_avg ~ 1,
  locations = soil_sp_B1,
  newdata = as(indiana_counties, "Spatial"),
  model = model_B1,
  block = c(20000, 20000)
)

#B2 Block Kriging
kriging_B2 <- krige(
  value_avg ~ 1,
  locations = soil_sp_B2,
  newdata = as(indiana_counties, "Spatial"),
  model = model_B2,
  block = c(20000, 20000)
)

#B3 Block Kriging
kriging_B3 <- krige(
  value_avg ~ 1,
  locations = soil_sp_B3,
  newdata = as(indiana_counties, "Spatial"),
  model = model_B3,
  block = c(20000, 20000)
)
```



The kriging results indicate that the uppermost B horizon (B1) has fewer counties with suitable soil pH compared to the deeper sublayers. This has implications for shallow-rooted crops, like strawberries, which primarily develop within the B1 layer and may therefore be limited by less optimal pH conditions. In contrast, deeper-rooted crops, such as fruit trees, which extend into the B2 and B3 sublayers, are likely to benefit from the more favorable pH conditions observed at these depths. Notably, the B3 sublayer shows the highest number of counties with suitable pH, suggesting that deeper soil layers may offer better agricultural potential in terms of pH suitability.

These results align with the variogram patterns, which showed lower variability and stronger

spatial correlation in B1, leading to more uniform but unfortunately, less favorable predictions. The higher variability seen in the B2 and B3 variograms is reflected in the kriging maps as greater spatial variability, which allowed kriging to identify more scattered pockets of suitability. This suggests that while deeper soils are generally more suitable, they are also spatially more variable, especially in southern counties.

```
# A tibble: 3 x 2
  depth_layer suitable_counties
  <chr>                   <int>
1 B1                         36
2 B2                         64
3 B3                         78
```

```
# A tibble: 79 x 4
   county_name B1    B2    B3
 * <chr>       <chr> <chr> <chr>
 1 Jay         Yes   Yes   Yes
 2 Delaware    Yes   Yes   Yes
 3 Wayne       Yes   Yes   Yes
 4 Shelby      Yes   Yes   Yes
 5 Grant       Yes   Yes   Yes
 6 Washington  Yes   No    Yes
 7 Tipton      Yes   Yes   Yes
 8 Whitley     Yes   Yes   Yes
 9 Howard      Yes   Yes   Yes
10 Blackford   Yes   Yes   Yes
# i 69 more rows
```